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SEPM Assignment 1

* Process Model in Software Engineering

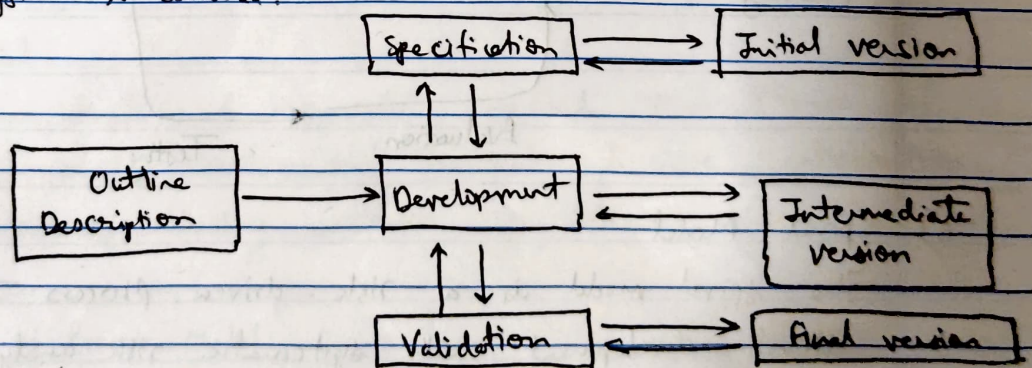
A process model in software engineering is a structured framework that defines the sequence of activities required to develop a software system. It provides guidelines for planning, designing, building, testing, & maintaining software in a systematic manner.

• Types of Process Models

1. Waterfall Model - The waterfall model is a linear sequential process where software development progress through fixed phases:
 - Requirement → Design → Coding & Unit Testing → Integration → operation & Systems Testing Maintenance

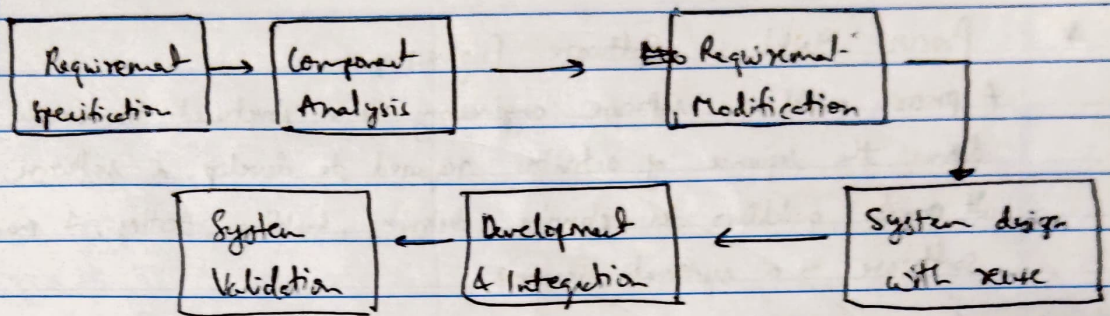
2) Evolutionary (~~Explain~~ Exploratory Model) :-

In this model, software is developed in multiple versions. The initial version provides basic functionality & subsequent version are enhanced based on user feedback until the final system is delivered.



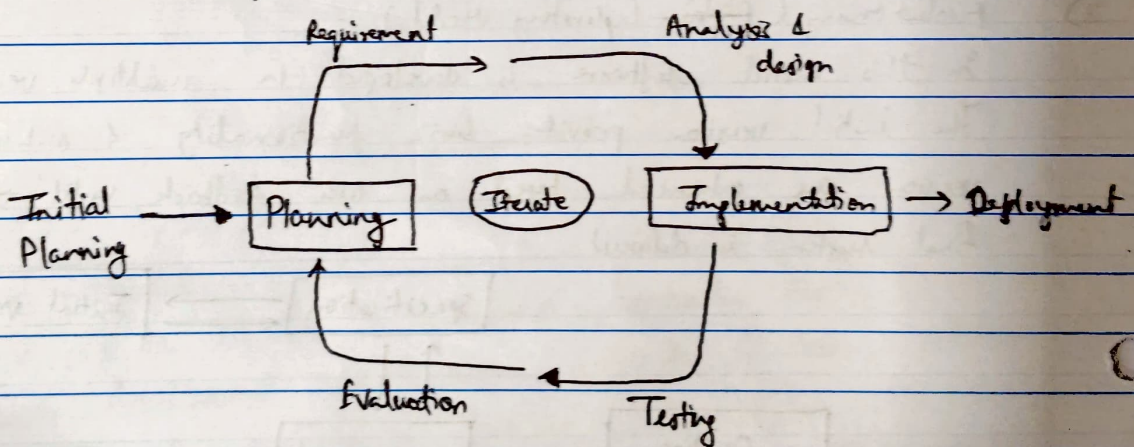
3) Component Based Software Engineering (CBSE) :-

CBSE focuses on building systems by reusing existing software components. The process includes requirement specification, component analysis, requirement modification, system design with reuse, development, integration & validation.



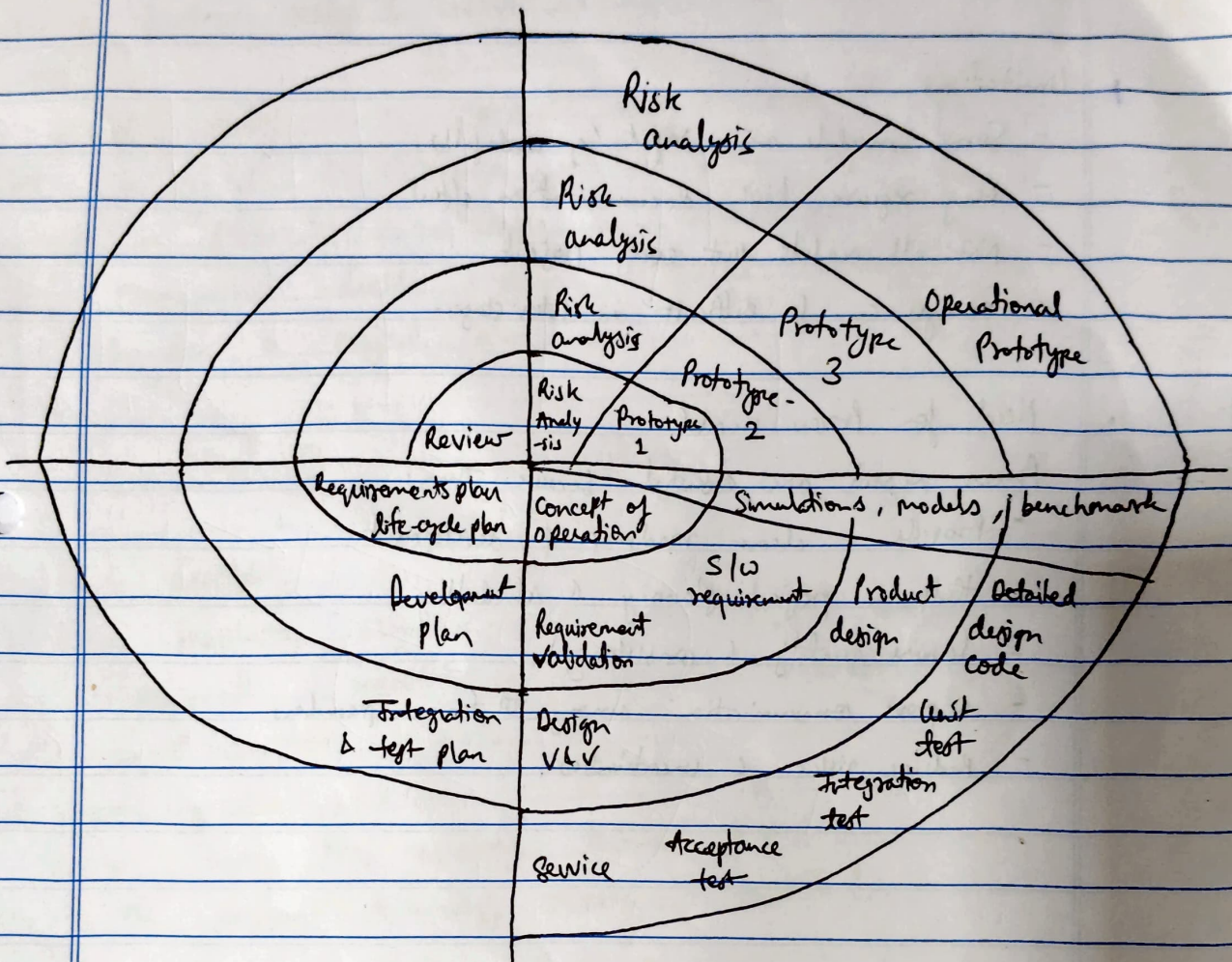
4) ~~Integration Model~~ Incremental Model

In the incremental model, software is developed & delivered in small increments. Each increment adds new functionality, & final system is achieved after all increments are completed.



5) Spiral Model

The spiral model is a risk-driven process that combines iterative development with systematic risk analysis. Each loop of the spiral planning, risk analysis, development & evaluation.



6) Agile Model

Agile is an iterative & incremental approach that focuses on rapid development continuous feedback, & close collaboration with customers. Working software is delivered frequently.

• Advantages of Process Models

- Provides structured development approach
- Improves end quality & reliability
- Facilitates better project control
- Helps in risk management.

- Limitations

- Some models are rigid (e.g. waterfall)
- May require high documentation effort
- Not all models suit every project
- Changes can be difficult in later stages.

- Need for Process Model

Process models are essential because they:

- Provide a clear roadmap for development
- Help in project planning & scheduling
- Ensure quality & consistency
- Improve communication among team members
- Reduce risks & uncertainties.